<u>AutoGenPrep 965 Tissue Protocol</u>

Adapted from [***AutoGen mouse tail tissue protocol***]

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Helpful Hints or Important Considerations

- <u>It is best to run this protocol for the first time with someone who is familiar using the AutoGenPrep 965</u>. Even if you follow this protocol exactly as written, the extraction can still fail due to user error or component failures. If this happens, it is helpful to have someone with the experience to troubleshoot the issue and restart the run.
- Tissue extraction is a three-day process. The first day involves preparing samples for overnight digestion. Day two involves using the AutoGenPrep 965 to extract DNA. Day three involves resuspending extracted DNA and preparing for long-term storage.
- You must sign up on our LAB AutoGen calendar (you need an SI account to access this calendar) to use the AutoGen. One AutoGen run of four plates (384 samples) takes about 3.5 hours, so try to sign up for a morning or afternoon run.

Materials & Reagents

- AutoGenPrep 965 Tissue DNA Kit
    - M1 Buffer
    - M2 Buffer
    - R3 Deproteinization
    - R4 DNA Precipitation
    - R5 DNA Wash
    - R8 Digestion
    - R9 DNA Resuspension
    - Proteinase K
    - Pipette tips
- Costar 96-well Polypropylene Storage Blocks [Corning#: 3960]
- Corning Axygen ImpermaMats, Chemical Resistant Silicone 96 Square Well Sealing Mat [Corning#:AM-2ML-SQ-IMP]
- Thermo Scientific Matrix 0.5 ml Alphanumeric tubes (Thermo Cat#: 4274) w/
- Thermo Scientific Matrix SepraSeal caps (Thermo Cat#: 4465)
- Thermo Scientific Matrix 0.5 ml 2D ScrewTop tubes, V bottom (Thermo Cat#: 3745)
- Lab tape
- Permanent marker

Protocol
Day 1 – Digestion
Notes: This protocol assumes you have tissue in 150 μl of M2 buffer in either a Costar or an Alpha-numeric plate (see Materials & Reagents for more details about plate types). We have a separate protocol for sampling tissue into plates with M2 buffer.

For Tissue in Costar plates

1. Add 150μL of M1 + proteinase K solution to each well in the Costar plate
   a. M1 + protK is good at 4°C for 1 month once the proteinase K has been added. If no bottles of M1 + protK (found in refrigerator 04-E1-211) are less than 1 month old, make a new mix
   b. Unopened bottles of M1 are on the shelves next to the AutoGenPrep. 50 or 100 mg vials of proteinase K are found in freezer (04-E1-205), on the left as you enter the extraction room (E1107). Add 100 mg of proteinase K into the full bottle of M1 buffer, secure with the cap, and invert to combine. Pour some M1 into the proteinase K bottle and back into the M1 to ensure all proteinase K powder has dissolved
      Note: *Be careful with proteinase K as it is an irritant and can cause discomfort if it contacts your eyes, nose, and skin*
   c. M1 + protK can be added to plates using the Rainin Liquidator 96 or a multi-channel pipette
   d. Label bottle of remaining M1 + protK with date made, and place in refrigerator 04-E1-211
2. Cover the Costar plate with a sanitized silicone mat (Corning#:AM-2ML-SQ-IMP). Place a discarded SepraSeal plate (a clear plastic plate with 96 small posts on one side) inverted onto the mat, so the posts are inside the indentations of the mat. Place a rigid plexiglass plate on top of this, and secure well with lab tape.
3. Leave costar plate overnight to incubate in a shaking incubator at 50 RPM and 56.5°C

For Tissue in alphanumeric plate

1. Add 150μL of M1 + proteinase K solution to each well in the alphanumeric plate
   1.1. M1 + protK is good at 4°C for 1 month once the proteinase K has been added. If no bottles of M1 + protK (found in refrigerator 04-E1-211) are less than 1 month old, make a new mix
   1.2. Unopened bottles of M1 are on the shelves next to the AutoGenPrep. 50 or 100 mg vials of proteinase K are found in freezer (04-E1-205), on the left as you enter the extraction room (E1107). Add 100 mg proteinase K into the full bottle of M1 buffer, secure with the cap, and invert to combine. Pour some M1 into the proteinase K bottle and back into the M1 to ensure all proteinase K powder has dissolved
      Note: *Be careful with proteinase K as it is an irritant and can cause discomfort if it contacts your eyes, nose, and skin*
   1.3. M1 + protK can be added to plates using the Rainin Liquidator 96 or a multi-channel pipette
   1.4. Label bottle of remaining M1 + protK with date made, and place in refrigerator 04-E1-211
2. Seal the alphanumeric plates with SepraSeals. The pre-PCR rooms at NHB and MSC both have an automated SepraSeal sealer.
   2.1. Do not attempt to use the SepraSeal automated sealer on plates that are not full, it will jam the sealer and you will not be able to retrieve your plate. If this happens, contact Jeff or Tripp. Partial plates can be sealed by hand using substantial pressure.
3. Leave alphanumeric plate overnight to incubate in a shaking bath at 50 RPM and 56.5°C

Day 2 – Extract DNA
If the samples were not digested in a Costar plate, then they will need to be transferred to one before they can be placed into the AutoGenPrep. If digestions occurred in Costar plates, go directly to step 6.

4. Transfer contents of each alphanumeric plate to a new Costar plate
   4.1. Label Costar plates
      4.1.1. Place a strip of lab tape on the middle of the long side of each Costar plate and label it with the plate name
      4.1.2. Make sure the tape does not extend to the edges of the plate.
      4.1.3. Because the plates are clear, the machine's sensors will not recognize that there are plates present without the tape
   4.2. Transfer contents using an 8-well multichannel pipette. **Do not use the Rainin Liquidator 96 to transfer digested material.**
   4.3. Remove SepraSeals for each column as you transfer digests. Do not remove all SepraSeals before starting transfer
   4.4. Take care when transferring the digest to not also transfer the tissue as it can clog the pipettes in the AutoGenPrep and cause the extraction to fail
5. Weigh the plates and add H_2O to any negative control wells to balance plate pairs
   5.1. Plates are run in pairs so balance plate 1 to plate 2 and plate 3 to plate 4. Try to balance to within 0.5 g. Unbalanced plates may cause the extraction to fail.
   5.2. If there are no empty wells, then evenly distribute M2 buffer among the wells until the plates are of equal weight
6. Label new Costar plates for extracted DNA.
   6.1. Place a strip of lab tape on the middle of the long side of each Costar plate and label it with the plate name
   6.2. Make sure the tape does not extend to the edges of the plate.
   6.3. Because the plates are clear, the machine's sensors will not recognize that there are plates present without the tape
7. Turn on the compressor located to the right of the AutoGenPrep. Listen for the hum as an indication it has been activated. *Note: If tank is already full, it will not turn on compressor until it needs to fill tank.*
8. Place all Costar plates on AutoGenPrep. *Figure 1* shows a diagram of how the plates should be placed, with empty Costar plates on the left and digests on the right.
9. Add tips to AugoGenPrep
   9.1. Remove plastic from 4 new tip boxes and place them as shown in *Figure 1*. Invert bottom box from package and place tip box into it. Place assembled tip box into Autogen. Make sure they are completely down, and all tips are at the same level (i.e. no tips are elevated above the rest).
   9.2. Make sure rows 3-8 on tip box 0 are filled. Only row 3 needs to be replaced between extraction runs, in which case use tips from a new box. Otherwise, replace entire tip box.

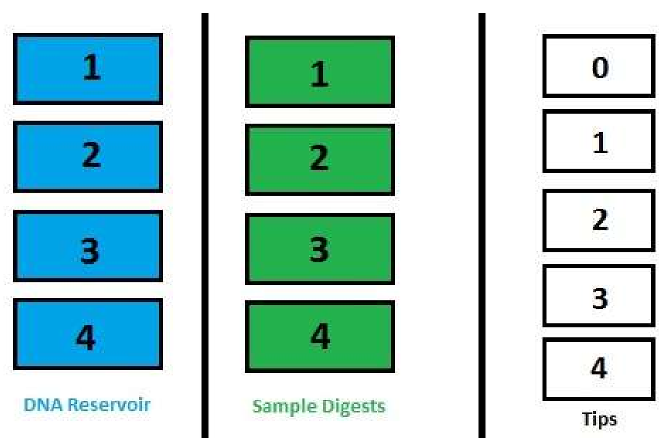

Figure 1. Plate and Tip layout in the AutoGenPrep 965. The Costar plates in the middle (seen in green) contain digests. The Costar plates on the left (seen in blue) are the empty plates from Step 6 and house the purified extract when the process is finished. Tip boxes are on the right. Tips boxes 1-4 correspond to samples 1-4 and must be replaced for each extraction run. Tipbox 0 is used for dispensing reagents to each plate, and its tips never encounter genetic material. Only tips in rows 3-8 are used, and only the tips in row 3 need to be replaced after each extraction run (they transfer reagent R3 - phenol).

10. Fill reagents for AutoGenPrep
    10.1. Get reagent reservoirs 4, 5, 6, and 8 from the drying rack. Fill these on table to left of AutoGenPrep
    10.2. Reservoir 3 will be under the fume hood facing the AutoGenPrep. Fill this reservoir while still in the fume hood.
    10.3. Fill the reservoirs up with the volumes of reagents indicated in the table below. Volumes are based on number of samples. 96 samples equal one plate. A modified version of this table can also be found in the AutoGenPrep mouse tail protocol in the booklet next to the extractor:

Reagent	Reservoir	Volume/Sample (mL)	Volume/Sample Number			
			96	**192**	**288**	**384**
R3 Deproteinization	3	0.18	42	60	77	94
R4 DNA Precipitation	4	0.20	45	64	83	102
R5 DNA Wash	5/6	0.45	69	112	155	198
R8 RNase	8	0.05	30	35	40	45

    10.4. Reservoir 4
        10.4.1. Fill with R4 DNA Precipitation (found in the yellow alcohol cabinet closest to the AutoGenPrep)
    10.5. Reservoir 5 & 6
        10.5.1. R5 DNA Wash (also found in the yellow alcohol cabinet closest to the AutoGenPrep).
        10.5.2. A third wash which uses Reservoir 7 can also be programmed but our standard protocol uses 2 washes.
    10.6. Reservoir 8
        10.6.1. R8 RNase (found in fridge 04-E1-211)
    10.7. Reservoir 3
        10.7.1. R3 Deproteinization (opened bottles are found in refrigerator 04-E1-201 on bottom shelf; unopened bottles are found in freezer 20-E1-205)
            10.7.1.1. Be careful when using phenol and avoid touching it as much as possible
            10.7.1.2. Do not touch any other surfaces with fingers that potentially touched R3
            10.7.1.3. Fill reservoir 3 under the fume hood and then immediately place the R3 back in the refrigerator
            10.7.1.4. Remove and replace gloves as soon as R3 has been returned to refrigerator
            10.7.1.5. Unopened bottles are found in freezer 04-E1-205, but once opened, should be stored on the bottom shelf in refrigerator 04-E1-201.
11. Check the waste bottles inside the cabinet in the AutoGenPrep. If the waste bottle is over a certain weight, the AutoGenPrep will not pass the system check or proceed to run. If more than 2/3 full, contact Jeff to see about disposing of the waste.
12. To activate the AutoGenPrep, look for the large, black ignition knob on the right side of the machine. Twist the knob counterclockwise to the reset position (green) then clockwise to the 12 o'clock position (red) to engage the machine
    12.1. If the machine does not turn on, then check that the emergency shutoff (the red button on the front of the AutoGenPrep) hasn't been pressed. If it has, pull it out with a twisting motion to reset and repeat the above step.
13. Access the computer to the right of the AutoGenPrep and click the AutoGenPrep icon located on the desktop
14. Click Standard protocol > OK > Mouse Tail > Automated Run > LAB General >
    14.1. Within the LAB profile a few tweaks can be made:
        14.1.1. Check that the sample # is correct (384 for 4 plates; 192 for 2 plates)
        14.1.2. If using Reservoir 7, check that the number of washes is set to 3

14.1.3. Make sure “blower” is turned off and “*next process*” = finish
14.1.4. Confirm changes to continue (changes from default are shown in yellow)

15. Start Systems Check
    15.1. If everything is OK, then click Start to begin the process. **This does not start the run, only the Systems Check.** This can take 5-10 minutes. I usually take this time to clean up (dispose of alphanumeric plates, discard tape or lids used in digestions, etc.)
    15.2. Once System check is over, you will have to press start again to start the run. You can periodically check to the status of a run by visiting this website: halite.si.edu. You will need to log in to use the website. The login is “lab”, and the password is LAB’s typical computer password. Both are on a printout near the AutoGenPrep.
16. Remove DNA reservoir plates
    16.1. Wrap plates with large Kimwipe
    16.2. Place covered plates under the hood on the far wall or into a drawer designated for your samples (see Jeff or Tripp). Allow to dry overnight.
17. Throw empty tip boxes away
18. Remove phenol-contaminated plasticware from AutoGenPrep. **Remove and dispose of gloves immediately after performing these steps**
    18.1. Put the plates that had contained digested tissue into the fume hood so the leftover fluid can evaporate
    18.2. Take the 3rd row of tips in tip box 0 and place them into the dry waste bag/container in the fume hood.
    18.3. Move reservoir 3 to the fume hood
        18.3.1. Empty any remaining phenol in reservoir 3 into the phenol waste container
        18.3.2. Rinse out reservoir 3 with the remnants of reservoirs 4,5,6, and 8, one at a time, disposing of the liquid into the phenol waste bottle each time.
        18.3.3. Cap the phenol waste container and return it to the cabinet
        18.3.4. Remove gloves and throw into dry waste bag/container in hood.
19. Wash reservoirs 4,5,6, and 8 at the sink behind the AutoGenPrep using copious amounts of water and leave to dry in the dish rack
20. Shut down the Autogen:
    20.1. Turn off compressor
    20.2. Turn off Autogen (right side of machine to green "off")
    20.3. Close Autogen program on laptop

Day 3 – Resuspend and Store DNA

If using extractions in the near future, rehydrate, otherwise, seal dried plates with foil or with heat sealer and place in a -20°C or -80°C freezer

21. Resuspend DNA in DNA plates with Resuspension buffer, (aka R9 Buffer).
    21.1. Use 50µl for small tissues, 100µl for big tissues e.g. fish, mammals, birds.
    21.2. You may use either the Rainin Liquidator 96 or a multichannel pipette to transfer buffer to extract wells
22. Seal resuspended DNA plates with foil and place into shaking bath, shake at room temperature for ~ 1 hr.
23. Transfer resuspended DNA to pre-scanned barcode tubes, reserving 5-10 µl in the costar plate as a working stock. It is important to scan barcode tubes (using the barcode plate scanner in the pre-PCR room) before filling with extracted DNA.

23.1. Seal working-stock plate with a new sheet of foil
23.2. Seal barcode tubes with new SepraSeals

24. Store barcode plate in your designated place for long-term DNA storage. Store working-stock plate in the freezer at your bench in post-PCR

References:

****AutoGen mouse tail tissue protocol reference*******